CCRM ENTERPRISE OPERATING SYSTEM • PRODUCT BRIEFING

THE AUTONOMOUS CRM & WORKFLOW REVOLUTION

Next-generation CRM ecosystem uniting Multi-Agent Swarm Intelligence, Visual Workflow Automation, Deep Financial Due Diligence, and Hybrid Offline-First Sync.

🍋 V1.11 LEMON | 🥝 V1.10 KIWI | 🍈 V1.9 JACKFRUIT | 🍊 V1.8 IMBE | 🫐 V1.7 HUCKLEBERRY | 🍇 V1.6 GRAPEFRUIT

CSTUDIOS ENTERPRISE ARCHITECTURE • EXECUTIVE SALES DECK • CONFIDENTIAL

Fragmented SaaS Stacks vs. Unified CCRM Architecture

Why traditional CRM subscriptions bleed capital, leak confidential client data, and break operations.

⚠ The Fragmented Enterprise Nightmare

- **Disjointed Tool Sprawl:** Companies pay separately for Salesforce/HubSpot, Zapier/Make, Gong/Voice AI, FinStat registries, and E2E test suites.
- **Outrageous Per-Seat Pricing:** A 25-person team easily exceeds $3,500/month with compounding third-party add-ons and tier lock-ins.
- **Sync Friction & Data Loss:** Webhooks fail silently, API rate limits throttle operations, and field changes vanish during browser refresh.
- **Sovereignty & Data Privacy Risks:** Client financial statements and confidential deals pass through dozens of unregulated cloud vendors.

⚡ The CCRM Unified Operating System

- **All-In-One Enterprise Core:** Full CRM, Visual Automation Canvas, Swarm AI Simulators, Financial Ledger, and QA Audit built into one engine.
- **One-Time / Predictable Cloud Hosting:** Zero per-seat artificial tax. Self-hosted on your own infrastructure or secure dedicated cloud instance.
- **Zero-Latency Offline-First Sync:** Instant UI responses, optimistic UI updates, checksum drift protection, and bulletproof background persistence.
- **100% Data Sovereignty:** Local MariaDB storage, zero third-party telemetry, encrypted secret vaults, and localized RAG grounding.

83% Reduction in annual software subscription costs compared to legacy CRM & iPaaS stacks

< 50ms Optimistic UI response time across all 12 modules with real-time state synchronization

COMPETITIVE MOAT

The 6 Pillars of CCRM Enterprise Superiority

Unmatched technological capabilities built directly into the core application engine.

1. Swarm AI (SAI) Simulation

Run predictive multi-agent market war-games. Rehearse price increases, contract disputes, and competitive retaliation before taking public action.

ZERO-DOCKER CLIENT ORCHESTRATION

2. Visual Workflow Engine

Full visual node-canvas automation. Connect database hooks, AI providers, conditional branching, and email triggers with zero external subscriptions.

N8N & MAKE PARITY

3. Deep Financial RAG

Direct live connection to RegisterÚZ and FinStat registries. Instant corporate balance sheets, solvency scoring, and EBITDA analysis in seconds.

NATIONAL REGISTRY INGESTION

4. Universal Favorites & Voice

Fast multi-entity pinning, quick-access drawer, and AI voice task capture. Talk on mobile, and CCRM automatically parses client, deadline, and priority.

SUB-SECOND NAVIGATION

5. Autonomous QA Auditor

Autonomous Playwright crawler with visual occlusion testing. Scans all 12 views in under 50 seconds to ensure 0 UI defects before deployment.

SELF-HEALING QUALITY GATE

6. Model Context Protocol (MCP)

Seamless enterprise connectivity. Supports 41 tools across OpenAI, Anthropic Claude, and Google Gemini with encrypted secret masking.

MULTI-LLM AGNOSTIC

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Hybrid Client-Server Enterprise Architecture

Combines the snappy speed of local browser runtime with robust server-side database durability.

1. Client UI & State

React 19, TypeScript, Tailwind CSS, Lucide Icons, and React Bits WebGL LightRays.

- Optimistic local mutation
- PWA mobile capabilities
- Screen wake-lock for SAI

2. Autonomous SAI Engine

Runs multi-agent turn loops in browser memory without heavy Python/Celery clusters.

- Linear RecSys feed ranking
- Dynamic per-sim table sharding
- ReAct strategic briefings

3. Backend & Sync Engine

High-speed PHP 8.3 REST & SSE server with real-time sync checksum validation.

- Automated sync-backend drift fix
- Encrypted secret vault (AES-256)
- Headless Craft CMS GraphQL sync

4. Storage & Data Fabric

MariaDB/MySQL relational storage paired with national corporate registries.

- RegisterÚZ & FinStat APIs
- Vector DB RAG embeddings
- Automated incremental backups

Data Flow & Resiliency Pipeline

User Interaction
Instant optimistic update (<16ms)

→

Checksum Engine
Calculates delta & version hash

→

sync.php Handshake
Verifies atomic lock & merges delta

→

Automation Hooks
Queues workflow triggers & notifications

→

RAG Index Refresh
Syncs vector DB context for AI agents

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The Fruit Release Evolution: Constant Innovation

A proven track record of rapid, reliable continuous delivery with zero regressions.

1.6 GRAPEFRUIT

Dynamic AI Dashboards

Introduced custom dashboard creation, WebGL LightRays login experience, and modernized Docker container foundation.

Released: Summer 2026

1.7 HUCKLEBERRY

Visual Automation Engine

Built n8n/Make-style drag-and-drop node workflow editor, cron task runner, and execution trace inspection debugger.

Released: Late Summer 2026

1.8 IMBE

QA Audit & Financial RAG

Deployed autonomous Playwright UI crawler, RegisterÚZ / FinStat RAG integration, and encrypted SecretInput state protection.

Released: August 2026

1.9 JACKFRUIT

Ledger & Drag-and-Drop

Engineered full financial ledger, recurring revenue engine, project drag-and-drop ordering, and system SMTP server.

Released: September 2026

1.10 KIWI

Swarm AI (SAI) Engine

Debuted multi-agent market simulation engine, Live War Room knowledge graph, Craft structure view, and pre-flight token guard.

Released: September 2026

1.11 LEMON (CURRENT)

Universal Favorites & MCP

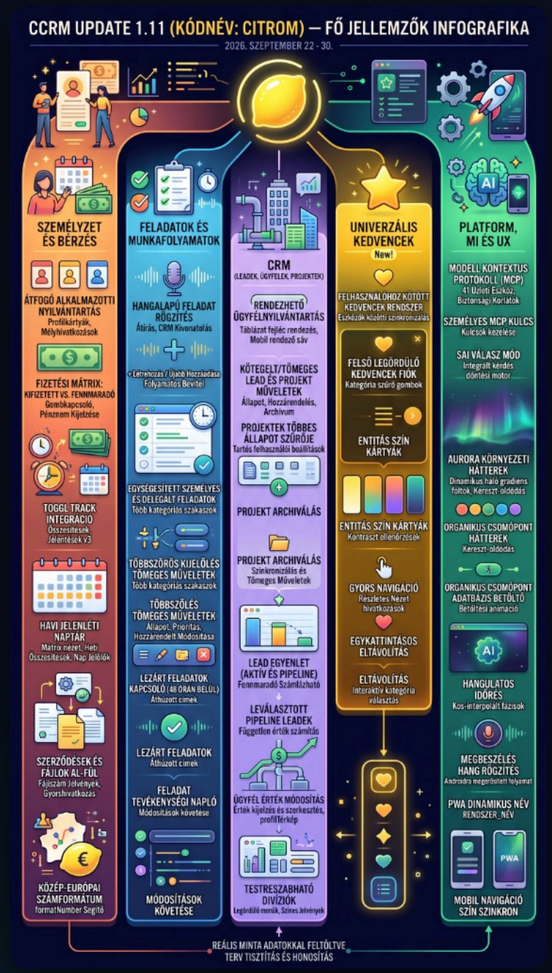
Universal Favorites drawer, team payroll & attendance matrix, voice task recorder, and 41-tool Model Context Protocol (MCP).

Active Release: Present

ACTIVE FLAGSHIP RELEASE

Update 1.11 (Codename: Citrom) — Universal Favorites & MCP

Revolutionizing daily productivity, team payroll management, and external agent integrations.



⭐ Universal Favorites & Sub-Second Drawer

- **Pin Anything:** Instant one-click bookmarking of Leads, Clients, Projects, Tasks, and Warehouse Items.
- **Top Slide-Out Drawer:** Accessible from any view with category pills, contrast-checked color chips, and keyboard navigation.
- **Cross-Device Profile Sync:** Favorites follow the authenticated user profile across mobile, tablet, and desktop.

💼 Team HR, Attendance & Payroll Matrix

- **Comprehensive Employee Roster:** Deep-linked profile cards, role management, and compensation rate tiers.
- **Monthly Attendance Calendar:** Matrix view, weekly summaries, day tags, and Toggl Track v3 integration.
- **Central European Number Formatting:** Native currency and decimal locale parsing across all financial figures.

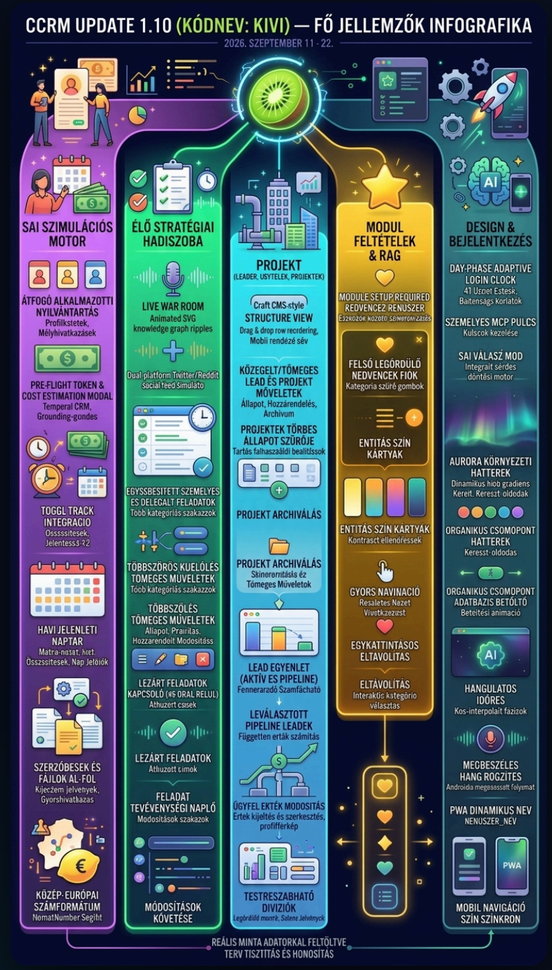
🎙️ Voice-to-Task & Model Context Protocol (MCP)

- **Android-Hardened Voice Capture:** Direct audio dictation with automated task name, deadline, and priority tagging.
- **Enterprise MCP Server:** 41 native business tools enabling autonomous external AI agents to perform audited CRM actions safely.

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Update 1.10 (Codename: Kivi) — Swarm AI & Live War Room

Autonomous multi-agent predictive simulation engine running entirely in modern browser runtimes.



🧠 Swarm Artificial Intelligence (SAI) Engine

- **Client-Orchestrated Multi-Agent Architecture:** Simulates dozens of autonomous personas without Python or Redis servers.
- **Linear RecSys Feed Algorithm:** Dynamic formula (0.4 Recency + 0.3 Popularity + 0.3 Relevance) drives agent information feeds.
- **Pre-Flight Cost Estimator:** Real-time token and USD cost forecasting for GPT-5.6 Luna / Terra before running a simulation.

⚔️ Live War Room & Knowledge Graph

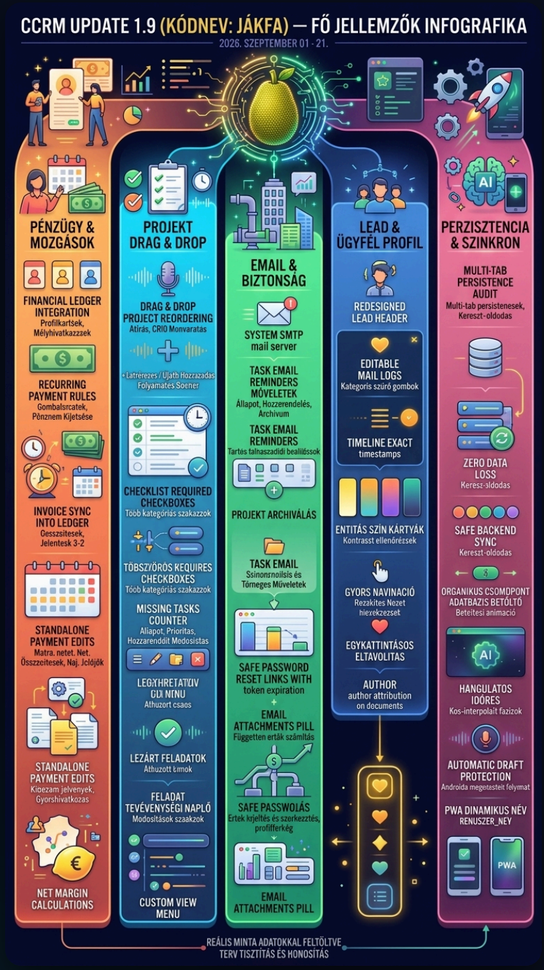
- **Animated SVG Entity Canvas:** Live visual ripples representing posting, retweeting, and counter-arguments between stakeholders.
- **Dual-Platform Feed Stream:** Cascading Twitter/X and Reddit mock feeds with real-time sentiment tracking and consensus bar.
- **ReAct Chief Intelligence Analyst:** Generates strategic executive briefing playbooks with actionable counter-measures.

📐 Craft CMS Structure View & Module Guards

- **Hand-Set Structure Order:** Drag-and-drop hierarchy view that preserves custom positioning during column sorts.
- **Setup Required State Guards:** Inactive modules gracefully guide users to missing API keys or DB connections.

Update 1.9 (Codename: Jákfa) — Unified Ledger & Drag-and-Drop

Robust financial cash flow intelligence, system-level communications, and multi-tab state persistence.



💰 Unified Financial Ledger & Recurring Cashflow

- **Invoicing Sync to Ledger:** Invoices generated in billing automatically feed the central cashflow ledger.
- **Standalone Payment Editing:** Edit specific monthly instances of recurring payments without breaking parent rule integrity.
- **Net Margin & Realized Cashflow:** Separates earned revenue from paid receipts with overdue receivables tracking.

✨ Project Drag-and-Drop & Checklists

- **Universal Drag Hook:** Smoothly reorder project rows or cards with animated visual drop guides.
- **Mandatory Task Checklists:** Project requirements tracking with a live "missing required tasks" counter badge.
- **Custom View Menu:** Save custom sort orders and column visibility preferences per user.

📧 System Mail Server & Persistence Audit

- **Dedicated System SMTP:** Task reminders, cron alerts, and password resets route through central server SMTP.
- **Multi-Tab Persistence Hardening:** Solved edge-case race conditions so background tab saves never overwrite active work.

Update 1.8 (Codename: Imbe) — QA Audit & Financial RAG

Autonomous UI exploration crawler, deep corporate registry synthesis, and encrypted credential masking.



🤖 Autonomous QA Exploration Crawler

- **Deep UI Crawler:** Traverses all 12 core application views, tabs, modals, and drawers in under 50 seconds.
- **Physical Occlusion Detection:** Uses DOM bounding geometry to autonomously detect when dropdowns are hidden behind overlays.
- **Zero-Regression Gate:** Deployments are strictly gated by automated Playwright test verification (`npm run test:qa`).

🏛️ RegisterÚZ & FinStat Financial RAG

- **Direct Corporate Intelligence:** Automatically injects company balance sheets, revenue, and profit history into AI agent context.
- **Actionable AI Error Codes:** Clear, human-readable codes (`ai_key_missing`, `company_not_found`) with direct settings shortcuts.
- **Warehouse Stock Awareness:** Autonomous agents retain real-time knowledge of stock levels, SKUs, and replenishment limits.

🛡️ SecretInput Encrypted Vault & Sync Drift Fix

- **State-Based Secret Masking:** Displays clean "Saved & encrypted" shield badges instead of confusing raw asterisk strings.
- **Accidental Clear Protection:** Warning modal with instant "Keep secret" rollback prevents accidental credential loss.

Update 1.7 (Codename: Huckleberry) — Visual Workflow Engine

Full-featured node-canvas automation engine replacing external Zapier and Make subscriptions.



⚡ Node-Based Workflow Canvas (n8n/Make Parity)

- **Drag-and-Drop Visual Graph:** Build complex multi-branch business logic with interactive horizontal bezier connection wires.
- **Collapsible Node Cards:** Expand to edit parameters, collapse to maintain clean canvas architecture with active centered ports.
- **Color-Coded Custom Dropdowns:** Module-themed selectors (Leads/Blue, Clients/Green, Tasks/Orange, System/Pink).

🔄 Real-Time Database Event Triggers

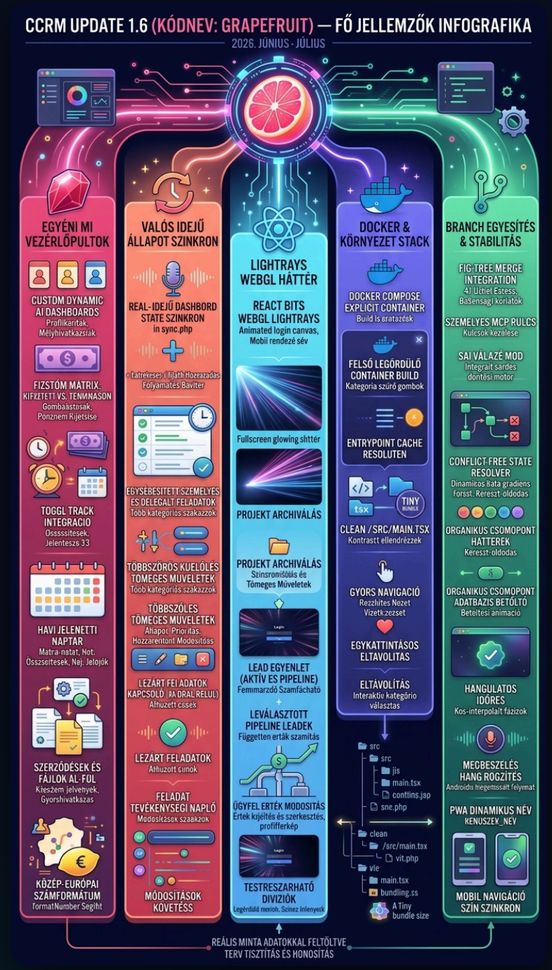
- **Native sync.php Hooks:** Fires workflows automatically on lead creation, stage transitions, client onboarding, and task updates.
- **Dynamic Variable Tag Popover:** Insert dynamic runtime variables (`{$trigger.name}`, `{$trigger.email}`) with one click.
- **Cron Task Runner:** Evaluates delay timers, scheduled batches, and recurring workflow loops securely via `api/cron.php`.

🔍 Visual Execution Debugger & Quick Actions

- **Run History Explorer:** Inspect the exact raw JSON input and output payload for any individual node across past executions.
- **Toolbox Quick-Actions:** Pinned trigger buttons in the global top bar with custom icons and color-coded themes.

Update 1.6 (Codename: Grapefruit) — Dynamic Dashboards & WebGL

User-customizable AI dashboards, cutting-edge WebGL visual shaders, and production Docker containerization.



📊 Custom Dynamic AI Dashboards

- **Custom Dashboard Engine:** Stored schema (`custom_dashboards`) supporting personalized widget layouts, icons, and colors.
- **Real-Time Sync Checksums:** Dashboard configurations are validated in the global sync loop with zero multi-client collision.
- **AI Prompt Customization:** Configure dedicated system prompts per dashboard to extract bespoke business insights.

✨ React Bits WebGL LightRays Login Experience

- **High-Performance 3D Shaders:** Replaced static canvases with dynamic fullscreen WebGL light rays originating from the left viewport.
- **Fluid Framerates:** Smooth 60 FPS GPU-accelerated rendering on both desktop workstations and mobile browsers.

🐳 Enterprise Docker & Build Optimization

- **Explicit Container Orchestration:** Clean `docker-compose.yml` stack with explicit `container_name: crm` bindings.
- **Vite Bundler Hot-Reload:** Restored root entry script compilation to `/src/main.tsx` eliminating cache bypass anomalies.

ENTERPRISE SCENARIO 01

Predictive Market War-Gaming: The 25% Pricing Rehearsal

How enterprise leadership simulates high-stakes strategic decisions before public execution.

1. The High-Risk Decision

A SaaS provider plans to raise enterprise retainer fees by 25%, introduce a 99.9% uptime SLA, and bundle a new WhatsApp developer API bridge.

- Risk of immediate client backlash
- Threat of rival competitor discounting
- Uncertainty in sales team rebuttals

2. Running the SAI Rehearsal

Leadership seeds CCRM's SAI engine with historical lost deal objections, CRM client transcripts, and competitor pricing profiles.

- 50 simulated agents debate over 3 rounds
- Twitter/Reddit feeds reflect community sentiment
- Consensus bar clusters into 4 primary factions

3. Strategic Executive Playbook

The Chief Intelligence Analyst delivers an immediate post-rehearsal strategic report with precise countermeasures.

- Grandfather existing clients for 6 months
- Highlight WhatsApp API cost savings ($400/mo)
- Pre-empt competitor FUD campaigns

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ENTERPRISE SCENARIO 02

Autonomous Client Onboarding & Contract Generation

Eliminating 95% of manual administrative busywork with visual drag-and-drop node logic.

Fully Automated End-to-End Deal Flow

1. Lead "Won"
sync.php triggers workflow when stage changes to WON

→

2. FinStat Due Diligence
Verifies IČO, VAT solvency, and official address automatically

→

3. Client & Project Created
Converts lead into formal Client and provisions Project Kanban

→

4. Invoicing & Contract
Generates initial deposit invoice and pushes to ledger

→

5. Email & Slack Alert
Notifies assigned project manager and welcomes client

0 min Human administrative time required to onboard won accounts and issue retainer agreements

100% Audit trail compliance with complete JSON execution trace inspection in visual debugger

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ENTERPRISE SCENARIO 03

Real-Time Corporate Financial Due Diligence via RegisterÚZ

Instant corporate balance sheet analysis and risk scoring directly inside client dossiers.

1. Automated Registry Lookup

Type or paste any Slovak or European company ID (IČO). CCRM connects to RegisterÚZ and FinStat APIs to pull official records.

- Full registered legal name & address
- Official tax IDs (DIČ, IČ DPH / VAT)
- Historical filings and balance sheets

2. AI Financial Synthesis

With one click, CCRM ingests raw corporate financial statements into the LLM reasoning pipeline.

- EBITDA, profit margin, and liquidity ratios
- Revenue trajectory over past 3 years
- Creditworthiness and insolvency flags

3. Risk-Adjusted Sales Strategy

Sales directors set contract payment terms based on empirical financial health rather than guesswork.

- Require upfront deposit for declining margins
- Offer annual retainer discount for high cash reserves
- Flag potential debt collection liabilities early

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Enterprise Security, Privacy & Data Sovereignty

Designed from the ground up for strict European data privacy, zero-leak credential storage, and RBAC.

Encrypted Secret Vault (SecretInput)

- **State-Based Secret Masking:** Passwords, API tokens, and private keys never render as plain text or ambiguous raw asterisks.
- **Server-Side Decryption Only:** Outgoing LLM completions and SMTP connections decrypt secrets purely in memory on the server.
- **Accidental Clear Protection:** Rollback warning modal protects against accidental token erasure.

Granular Role-Based Access Control (RBAC)

- **15+ Discrete Permission Nodes:** Control view and edit rights for Financials, AI Settings, Client Lists, and User Rosters.
- **View Inactivity Guards:** Users without permission see clear fallback cards rather than broken runtime exceptions.
- **Personal vs. System Scopes:** Clear separation between private personal settings and company-wide configurations.

GDPR & European Data Sovereignty

- **100% Private Cloud / On-Prem:** Host on your own EU servers or isolated VPC; zero US vendor data transfer requirements.
- **Zero Model Training:** Your client communications, financials, and transcripts are never shared with AI model trainers.
- **Right to be Forgotten:** Atomic customer deletion removes all associated notes, leads, and simulation records.

Automated QA Crawler Quality Gate

- **Autonomous UI Regression Testing:** Headless Playwright sweeps simulate authentic user journeys before code commits.
- **Physical Occlusion Inspection:** Ensures z-index conflicts and hidden interactive elements are flagged immediately.
- **Continuous Integrity:** Every build passes strict static type verification (`tsc --noEmit`) and integration suites.

Total Cost of Ownership (TCO) & ROI Analysis

Comparing annual software expenditures for a typical 25-person high-growth business team.

Software Category / Capability	Traditional SaaS Tool	Typical 25-Seat Cost / Year	CCRM Integrated Platform	CCRM Advantage
Core CRM & Pipeline Management	HubSpot / Salesforce Sales Cloud	$15,000 – $24,000	✓ Included in Core	Zero per-seat markup
Workflow Automation (iPaaS)	Make.com / Zapier Enterprise	$3,600 – $7,200	✓ Included Node Canvas	Unlimited internal runs
Corporate Due Diligence / Registries	FinStat Corporate + RegisterÚZ API	$1,800 – $3,000	✓ Built-in Direct API RAG	Seamless client dossier link
AI Strategy & Market Simulation	Dedicated AI Consultancies / Custom	$18,000 – $35,000	✓ Native SAI Engine	Direct CRM groundings
Continuous QA & Regression Testing	BrowserStack / Datadog Synthetics	$4,800 – $9,600	✓ Built-in QA Crawler	Zero additional infra cost
TOTAL ESTIMATED ANNUAL COST	**Fragmented SaaS Stack**	**$43,200 – $78,800 / year**	**CCRM Platform**	**Over 80% Capital Savings**

$52,000+ Average annual net savings achieved by consolidating SaaS operations into CCRM

4.2x Faster time-to-value with pre-integrated workflows, registries, and real-time state sync

SPEED TO VALUE

Frictionless Turnkey Deployment & Onboarding

Go live in days rather than quarters with our streamlined migration and validation pipeline.

Phase 1: Deploy

DAY 1 – 2

Rapid deployment on your private server, Docker cluster, or managed cloud instance.

- Automated DB migrations
- Docker Compose startup
- SSL & Domain configuration

Phase 2: Migrate

DAY 3 – 4

Seamless data import from legacy CRMs, CSV spreadsheets, and financial ledgers.

- Lead & client records import
- Historical timeline sync
- Company IČO data enrichment

Phase 3: Automate

DAY 5 – 7

Configure custom business workflows, AI model keys, and automated team notification rules.

- Node workflow activation
- OpenAI / Claude API link
- System SMTP configuration

Phase 4: Scale

DAY 8+

Team training, mobile PWA rollout, and continuous updates via Craft CMS release channel.

- Zero-downtime updates
- Automated QA gate checks
- Enterprise dedicated support

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THE NEXT EVOLUTION OF ENTERPRISE CRM

Ready to Elevate Your Operations?

Experience the unmatched speed, predictive power, and workflow automation of CCRM in an interactive private demonstration.

EXECUTIVE CONTACT

Erik Csefalvay

Managing Director, CStudios

erik@cstudios.sk

LIVE DEMONSTRATION

Custom War Room Simulation

Experience SAI live on your industry's specific challenges and competitors.

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